

#d.u.m.b.a.s.s. — Database & Universal Memory Bank Across Split Services

System Overview & Motto

"It isn't 'dumbass-proof' if it hasn't been '#d.u.m.b.a.s.s.' proven."

#d.u.m.b.a.s.s. is the unified database and memory backbone powering the distributed cognitive agent architecture across **Node Alpha** (Motorola RAZR Ultra 2025 on Snapdragon 8 Elite), **Node Beta** (NVIDIA Jetson Orin Nano Super), and edge sensory daemons (**Æsc** & **Æyre** alongside **Horizons UI**).

It provides split-service state management, decoupling fast on-device episodic caches from heavy centralized vector indexing and immutable audit ledgers.

Component Matrix

Subsystem	Primary Host / Role	Core Technologies	Primary Artifacts
sqlite/	Node Alpha (Termux / Æsc)	SQLite 3 WAL, C++ / Python bindings	schema.sql, README_SQLITE.md
postgres/	Node Beta (Jetson Orin Nano)	PostgreSQL 16+, pgvector, pgcrypto	migrations_v1.sql, README_POSTGRES.md
mem0/	Edge & Node Beta (P2P Sync)	Mem0 episodic tracking engine	rolling_habits_schema.json, mem0_session_config.yaml
ob1_protocol/	Federated MCP Layer	Open Brain (OB1) MCP Server protocol	ob1_mcp_spec.md
omniroute/	Data Routing & Memory Extraction Layer (Port 20128)	Proxy router, RTK token compression, async memory tap to	omniroute_config.yaml

Subsystem	Primary Host / Role	Core Technologies	Primary Artifacts
		Reasoning Bank & mem0	
<code>reasoning_bank/</code>	System Flywheel & Recovery	State checkpointing & RLVR grading	<code>active_execution_paths.json</code> , <code>recovery_protocol.md</code>
<code>continual_harness/</code>	Adaptive Runtime Layer	Reset-free online adaptation	<code>continual_harness_policy.md</code>

OmniRoute: The Data Routing & Memory Extraction Layer

OmniRoute (listening on `http://localhost:20128/v1`) functions as a dual-role gateway in the #d.u.m.b.a.s.s. architecture:

- **Data & Inference Routing:** Dynamically directs prompts across the hardware mesh (local Hexagon NPU → Jetson Orin Nano → cloud frontier models) while applying RTK token compression (15%–95% savings).
- **Asynchronous Memory Extraction:** Every prompt and completion passing through the gateway is asynchronously tapped. Tool invocations, successful reasoning steps, and failure trajectories are captured and dispatched into the **Reasoning Bank** and **mem0** without imposing latency on the inference loop.

Core Memory Laws

1. **Law of Sensory Immutability:** Raw documentation, historical logs, and base prompts are immutable. Changes occur only through additive versions or layered refinement deltas.
2. **Law of Zero-Loss Resumption:** Every multi-step agent trajectory is checkpointed to `active_task_state` and `active_execution_paths.json`. An interrupted session resumes from its latest verified step without repeating side-effects.
3. **Law of Local Locality with Global Discovery:** Subsystems operate against their own high-speed local caches and manifests, reaching into the centralized PostgreSQL store only when cross-cutting retrieval or deep embedding search is required.
4. **Law of the Sealed Red Auditor:** Auditing and validation remain strictly decoupled. Trajectories flagged with errors or hallucinated paths are quarantined with full error traces, preserving system integrity.